$\varphi: G \to H$ mapping

$\varphi(ab) = \varphi(a)\varphi(b) \quad \forall a, b \in G$

$\varphi$ is a homomorphism from $G$ to $H$

$\varphi: G \to G$ endomorphism

$G \to G$ iso $\Rightarrow$ automorphism

① $G \xrightarrow{\varphi} G' \xrightarrow{\psi} G''$

$\varphi, \psi$ hom $\Rightarrow \psi \circ \varphi$ is hom

iso $\Rightarrow$ iso

② $\varphi: G \to H$ iso, so is $\varphi^{-1}$

Pf: $\varphi^{-1}(h_1h_2) = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$, $\forall h_i \in H$

Since $\varphi$ is injective.

$\varphi^{-1}(h_1h_2) = \varphi^{-1}(h_1)\varphi^{-1}(h_2)$

$\Leftrightarrow \varphi(\varphi^{-1}(h_1h_2)) = \varphi(\varphi^{-1}(h_1)\varphi^{-1}(h_2))$

$\underset{h_1h_2}{\shortparallel} \quad = \varphi\varphi^{-1}(h_1)\varphi\varphi^{-1}(h_2)$

$= h_1h_2.$

Cor $Aut(G) = \{\varphi \mid \varphi$ is automorphism of $G\}$

$\leq Sym(G)$

$\forall a \in G.\ I_a: G \to G$

$x \mapsto axa^{-1}$

$I_a(xy) = axya^{-1} = axa^{-1}aya^{-1} = I_a(x)I_a(y)$

$I_a I_{a^{-1}} = id_G = I_{a^{-1}} \circ I_a \Rightarrow (I_a)^{-1} = I_{a^{-1}}$

$I_a \circ I_b = I_{ab}$

$\{I_a \mid a \in G\} \leq Aut(G)$

$I: G \to Inn(G)$

$a \mapsto I_a$

$\varphi(ab) = I_{ab} = I_a \circ I_b = \varphi(a)\varphi(b)$

$Ker\varphi = \{a \in G \mid \varphi(a) = I_a = id_G\}$

$= \{a \in G \mid I_a(b) = b \ \forall b \in G\}$

$aba^{-1} = b \Leftrightarrow ab = ba$

$= C(G)$ is called center of $G$

$Ax = b \quad \eta_0 + X$

prop 1.5.2 $f: G \to H$ hom

① $Ker f \trianglelefteq G$, $Im f \leq H$

② $f^{-1}(f(a)) = \{x \in G \mid f(x) = f(a)\}$

$= aKer f \in G/Ker f$

③ $f$ induces a bijection from

$G/Ker f \to Im f$

$\varphi: aKer f \to f(a)$

$x \in f^{-1}(f(a)) \Leftrightarrow f(x) = f(a).$

$\Rightarrow f(a)^{-1}f(x) = f(a^{-1}x) = e_H$

$\therefore a^{-1}x \in Ker f \Rightarrow x = a(a^{-1}x) \in aKer f$

$\forall ax \in aKer f \quad x \in Ker f$

$f(ax) = f(a)f(x) = f(a)e_H = f(a)$

③ i) $\varphi$ is well-defined.

$aKer f = bKer f \Rightarrow a = bx.\ x \in Ker f.$

$\Rightarrow f(a) = f(bx) = f(b)$

$\begin{cases} \varphi(aKer f) = f(a) \\ \varphi(bKer f) = f(b). \end{cases} \Rightarrow f(a) = f(b)$

$Im\varphi = Im f$

$\varphi(aKer f) = \varphi(bKer f)$

$\Rightarrow aKer f = bKer f$

$(aKer f)(bKer f) \xrightarrow{\varphi} \varphi(aKer f)\varphi(bKer f)$

$abKer f \xleftarrow{\varphi^{-1}} \varphi(abKer f) = f(a)f(b) = f(ab)$

$G/Ker f = \{aKer f \mid a \in G\}$

$aKer f \cdot bKer f = abKer f$

$e_H Ker f = Ker f \quad (aKer f)^{-1} = a^{-1}Ker f$

If $H \leq G$. $G/H = \{aH \mid a \in G\}$.

$(aH)(bH) \overset{?}{=} abH.$

is not well-defined.

$aH = a'H \quad bH = b'H.$

and $abH \neq a'b'H$

$G = S_3 \quad H = \{(1), (12)\}$

$\begin{cases} (123)H = (123)(12)H \\ (23)H = (23)(12)H \end{cases}$

Suppose $H \lhd G$. Then $G/H := \{aH \mid a \in G\}$

multiplication $(aH)\cdot(bH) := (ab)\cdot H$

is well-defined

$\begin{cases} aH = a'H \\ bH = b'H \end{cases} \Rightarrow \begin{cases} a' = ax \\ b' = by \end{cases}$ $x, y \in H$.

$a'b' = axby = ab(b^{-1}xb)y \in abH$

$\left.\begin{matrix} H \lhd G \\ x, y \in H \end{matrix}\right\} \Rightarrow b^{-1}xby \in H$

$a'b' \in a'b'H \cap abH \Rightarrow a'b'H = abH$

$G/H$ is called a quotient group of $G$ w.r.t. $H$

$\pi: G \to G/H$ canonical mapping

$a \to aH$ natural mapping

$\pi(ab) = abH = (aH)(bH) = \pi(a)\pi(b)$

Thm 1.5.1. $N \leq G$, $N \lhd G \Leftrightarrow \exists$ hom $f: G \to H$ s.t. $\operatorname{Ker} f = N$

Pf: $\pi: G \to G/N$

$\operatorname{Ker}\pi = \{a \in G \mid \underset{a \in N}{aN = N}\}$

Example: $S_n \to GL(n, F) \xrightarrow{|\cdot|} (F^*, \cdot)$

$\sigma \to (e_{\sigma(1)} \cdots e_{\sigma(n)})$

$\operatorname{sgn}(\sigma) = \det(e_{\sigma(1)} \cdots e_{\sigma(n)})$

$\operatorname{Im}\operatorname{Sgn}(\sigma) = \{\pm 1\} \cong \mathbb{Z}_2$.

$\operatorname{Ker}\operatorname{sgn} = \{\sigma \in S_n \mid \operatorname{sgn}(\sigma) = 1\} \lhd S_n$

$\operatorname{Sgn}(\sigma) = 1$

$S_n = \langle \sigma_1, \sigma_2 \cdots \sigma_{n-1} \rangle$ $\sigma_i = (i, i+1)$

$\sigma = \sigma_{i_1}\sigma_{i_2} \cdots \sigma_{i_r}$

$\sigma_{i_1} \cdots \sigma_{i_r} \in \{\sigma_1, \cdots \sigma_{n-1}\}$

$\operatorname{Sgn}(\sigma) = \operatorname{sgn}(\sigma_{i_1}) \cdots \operatorname{sgn}(\sigma_{i_r})$

$= (-1)^r$

$\operatorname{sgn}(\sigma) = 1 \Leftrightarrow r$ is even

$S_n/A_n \cong \{\pm 1\}$

$[S_n : A_n] = 2$.

Example $\varphi: F^n \to F^m$ linear transformation

$\varphi\left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}\right) = A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ $A = (a_{ij})_{m \times n}$

$\operatorname{Ker}\varphi = \left\{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in F^n \;\middle|\; A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = 0\right\}$

$A\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ 有解 $\in \operatorname{Im}\varphi$. $\eta_0$

$\varphi(\eta_0) = A\eta_0 = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$

$\varphi^{-1}(\varphi(\eta_0)) = \varphi^{-1}\left(\begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}\right) = \eta_0 + \operatorname{Ker}\varphi$.

$G/\operatorname{Ker} f \cong \operatorname{Im} f$

$V/W = \{v + W \mid v \in V\}$

$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W$

$k(v + W) = kv + W$

$\dim V/W = \dim V - \dim W$

Thm 1.5.3.

$f: G \to H$ an epimorphism

① $L \leq H$, $f^{-1}(L) = \{a \in G \mid f(a) \in L\} \leq G$

i.e. $L = \{e\}$

$f(f^{-1}(L)) = L$.

② $\operatorname{Ker} f \leq f^{-1}(L)$ $\forall L \leq H$

$f^{-1}(L)/\operatorname{Ker} f \cong L$

③ $\forall L \lhd H$ $G/f^{-1}(L) \cong H/L$

(1) $f^{-1}(L) \neq \phi$ ($e \in f^{-1}(L)$)

$\forall a, b \in f^{-1}(L) \Rightarrow a^{-1}b \in f^{-1}(L)$

$f(a), f(b) \in L \leq H \Rightarrow f(a)^{-1}f(b) \in L$

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad \overset{\shortparallel}{f(a^{-1}b)}$

$f(f^{-1}(L)) = L$

② $\{e\} \leq L$ $\underset{\operatorname{Ker} f}{\underset{\shortparallel}{f^{-1}(\{e\})}} \leq f^{-1}(L)$

$\varphi: f^{-1}(L) \to L$

$x \to f(x)$

$\operatorname{Im}\varphi = f(f^{-1}(L)) = L$

$\mathrm{Ker}\,\varphi := \mathrm{Ker} f$

$\therefore f^{-1}(L)/\mathrm{Ker}\varphi \cong \mathrm{Im}\,\varphi$

$f^{-1}(L)/\mathrm{Ker} f$

$\psi: G \to H/L$

$a \mapsto f(a)L$

$\psi(ab) = f(ab)L = f(a)f(b)L = \psi(a)\psi(b)$

$\mathrm{Im}\,\psi = \mathrm{Im} f/L = H/L$

$\mathrm{Ker}\,\psi = \{a \in G \mid f(a)L = L\} = \{a \in G \mid f(a) \in L\} = f^{-1}(L)$ 有 $e \in L$

$G/\mathrm{Ker}\psi \cong \mathrm{Im}\,\psi = H/L$

$= G/f^{-1}(L)$

Cor $N \lhd G$ $\pi: G \to G/N$

$N \leq \pi^{-1}(L) \leq G$

$\pi(\pi^{-1}(L)) = L = \pi^{-1}(L)/N$

$N \leq \pi^{-1}(L) \leq G$

example $\mathbb{Z}/12\mathbb{Z}$, $H$ is a subgp of $\mathbb{Z}/12\mathbb{Z}$

$H = H'/12\mathbb{Z}$

$12\mathbb{Z} \subseteq H' \leq (\mathbb{Z},+)$

$H' = n\mathbb{Z}$ $n\mathbb{Z} \supseteq 12\mathbb{Z}$

$n \mid 12 \Rightarrow n = 1, 2, 3, 4, 6, 12$

(2) $H/N \lhd G/N$

$\Rightarrow G/\pi^{-1}(H/N) \cong (G/N)/(H/N)$

$= G/H$. ($H \supseteq N$, $\pi^{-1}(\pi(H)) = H$)

核理:

第三同构定理: $H \lhd G$, $N \lhd G$ 且 $N \leq H$ 则 $(G/N)/(H/N) \cong G/H$

Proof: $G/H = G/\pi^{-1}(H/N)$

只需证: lemma1. $\varphi: G \to H$

若 $L \lhd H$ 则 $G/\varphi^{-1}(L) \cong H/L$

lemma2: 若 $H \lhd G$ $N \lhd G$ $N \leq H$ 则 $H/N \lhd G/N$

proof of lemma 1:

构造 $\psi: a \mapsto \varphi(a)L$ 则 $\mathrm{Ker}\,\psi = \varphi^{-1}(L)$.

$\mathrm{Im}\,\psi = H/L$.

proof of lemma 2:

构造 $\psi: G/N \to G/H$

$gN \mapsto gH$

$\mathrm{Ker}\,\psi: H/N$.

$H/N \lhd G/N$

$\Rightarrow$ 综上 $G/H = G/\pi^{-1}(H/N) \cong (G/N)/(H/N)$

$\mathbb{Z}/12\mathbb{Z} \cong \mathbb{Z}_{12}$

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n = \{\bar{0}, \bar{1}, \bar{2}, \cdots, \overline{n-1}\}$

$\pi: \mathbb{Z} \to \mathbb{Z}_n$ $\pi$ epic

$k \to \bar{k}$

$\mathrm{Ker}\,\pi = n\mathbb{Z} \Rightarrow \mathbb{Z}/\mathrm{Ker}\pi \cong \mathbb{Z}_n = \mathrm{Im}\,\pi$.

$\mathbb{Z}/n\mathbb{Z}$: $k + \mathbb{Z}_n \to \bar{k}$

$\dim(V_1+V_2) = \dim V_1 + \dim V_2 - \dim(V_1 \cap V_2)$

$H_1, H_2 \leq G$

$H_1H_2 := \{ab \mid a \in H_1, b \in H_2\}$

Prop. $H_1H_2 \leq G \Leftrightarrow H_1H_2 = H_2H_1$

Pf: $\Rightarrow \forall h_1h_2 \in H_1H_2 \Rightarrow h_1h_2 \in H_2H_1$

$\therefore h_2^{-1}h_1^{-1} \in H_2H_1$

$(h_1h_2)^{-1} \in H_1H_2$

$\therefore H_1H_2 \subseteq H_2H_1$.

$\forall h_2h_1 \in H_2H_1 \Rightarrow h_2h_1 = (eh_2)(h_1e) \in H_1H_2$. ($eh_2 \in H_1H_2$, $h_1e \in H_1H_2$; $e \in H_1$, $e \in H_2$)

$\Leftarrow$ $H_1H_2 = H_2H_1$